

AYUSH PATEL

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EDUCATION

McMaster University

Bachelor of Engineering in Software Engineering (CO-OP) - GPA: 3.9/4

Hamilton, ON

Expected Graduation: May 2027

- Coursework: Software Design I & II | Software Architecture | Object-Oriented Programming | Data Structures & Algorithms | Databases | Software Testing | Requirements Engineering & Security Considerations | Computer Architecture | Digital Systems | Discrete Mathematics I & II | Statistics

Awards: NSERC USRA | McMaster Award of Excellence | Frank Sherman Scholarship

EXPERIENCE

Software Developer Intern

SOTI

Sept 2025 - Aug 2026

Mississauga, ON

- Shipped full-stack features for SOTI MobiControl end-to-end using **Angular** for UI, **C#/.NET** for REST API development, and **SQL Server** database within an **Agile** team using **Jira** for project management
- Diagnosed and resolved **3+ critical production bugs** in user/session management, including a multi-year race-condition bug that resulted in significant performance improvements
- Drove the development of an autonomous troubleshooting agent from POC to production-ready code, building and validating agent/**MCP server** architecture and leading demos for senior leadership
- Migrated legacy production services to **.NET 10** to fix security vulnerabilities, resolving broken **Jenkins CI/CD** pipelines along the way

Software Research Intern

McMaster Center for Software Certification (McSCert)

May 2025 - Aug 2025

Hamilton, ON

- Developed an **AI-driven privacy transparency system** for mobile health apps, featuring a novel embedded UI that alerts users to data collection events with real-time, easy-to-understand explanations of privacy risks
- Engineered a full-stack system using **React Native**, **Express.js**, and **Firestore**, abstracting key components like sensor data, PHI storage, and LLM services to create a modular and scalable architecture.
- Ran a controlled experiment to evaluate AI-generated privacy explanations, finding a moderate positive correlation ($r=0.515$) between explanation length and logical consistency

PROJECTS

Scleroderma App | React Native, Firebase

Jan 2025 - Present

- Building a cross-platform mobile app using **React Native** and **Firebase**, delivering financial and mental health resources to scleroderma patients, reaching **1000+ users**

MovieWhiz | React Native, Express.js, RESTful API, Jest, UML, Firebase

Jan 2025 - May 2025

- Led the design and development of MovieWhiz, an **Android mobile application** that helps users identify forgotten movies using multiple expert systems including LLM's, soundtrack recognition and comprehensive movie databases.
- Managed the **full software development lifecycle**, including requirements gathering, system architecture design, UML diagrams, implementation, and testing, ensuring project coherence and quality.
- Developed a **RESTful API** backend using **Express.js** and tested it using **Jest** and Supertest, ensuring robust and reliable application functionality as well as data management using **Firestore**
- Designed for **scalability and extensibility**, enabling seamless integration of additional expert systems, LLM's and movie databases.

RescueMission | Java, Apache Maven, JUnit, UML, GitHub

Jan - Mar 2024

- Designed and implemented a **rescue drone control program** to locate stranded individuals and identify optimal rescue points, integrating efficient battery management algorithms to maximize operation time.
- Applied **SOLID principles** and **GoF design patterns** to build a scalable, maintainable system, leveraging object-oriented design and robust unit testing with **JUnit** to ensure functionality and reliability

TECHNICAL SKILLS

Languages: C#, Python, Java, SQL, TypeScript

Developer Tools and Frameworks: .NET, Git, React Native, Angular, MCP, Express.js, Node, Apache Maven, JUnit, SQL Server, Firebase, Jest, UML, Docker, Kubernetes

PROFESSIONAL CERTIFICATIONS

AWS Certified Cloud Practitioner (May 2025)

Introduction to Containers w/ Docker, Kubernetes & OpenShift (May 2025)